MIT SCHOOL OF COMPUTING
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MIT ART, DESIGN & TECHNOLOGY UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

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Subject: python Roll No.: 63 Exp. No.:
Name of the Experiment: Assignment - 1

Performed on:
Submitted on:

Marks	Teacher's Signature with date
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Introduction to python programming

Assignment - 1

1. list the key features and advantages of python. Also mention any 2 IDEs or editors used for program development.

Key features & Advantages of python

1. High level language - Easy to read and write
2. Interpreted language - Executes code line-by-line, easy debugging
3. Simple syntax - uses indentation insted of braces
4. Dynamic typing - No need to declare data types
5. Object-oriented - supports classes and objects
6. Open source - free to use and modify
7. Portable - Runs on windows, linux, Mac.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

8. large libraries - Numpy, Pandas, Matplotlib, etc.
9. Embeddable & extensible - can be integrated with other languages.

Any two python IDES

- Pycharm
- Jupyter Notebook

2. Explain the history and evolution of python. Why is python popular for beginners and industry?

History & evolution

- Python was created by Guido Van Rossum in 1989.
- Released in 1991
- Named after Monty Python
- Designed for simplicity and readability
- Python 3.x is the latest and widely used version.

Why python is popular

- Easy syntax for beginners
- used in AI, ML, Web Development, Data Science
- Huge community support.
- used by companies like google, Netflix, Instagram.
- faster development with fewer lines of code.

3. python program to acept 2 numbers and perform arithemtic operations ?

```
a= float (input ("Enter first numbers : "))
b= float (input (" Enter second number : "))

print ("Addition = ", a+b)
print ("Subtraction = ", a-b)
print (" Multiplication = ", a*b)
print (" Division = ", a/b)
```

Output

```
Enter first number : 10
Enter second number : 5
Addition = 15
Subtraction = 5
Multiplication = 50
Division = 2.0
```

4. Analyze the following python code and identify the data types of each variable. Also explain the output.

```
a = 10
b = 5.5
c = 3+4j
d = True
print (a+b, type(c), d)
```

Data types

- a → int
- b → float
- c → complex
- d → bool

Output explanation

- a+b = 15.5
- type (c) → <class 'complex'>
- d → true

5. Evaluate the use of relational, logical and bitwise operators in python. Justify with suitable expmples where each type of operator is most appropriate.

- Relational operators :
  used for comparision, used in conditional & Decision making

```
a = 10
b = 5
print (a > b)
```

- Logical operators :
  used to combine conditions
  used in authetication & validation.

```
x = True
y = False
print (x and y)
```

- Bitwise operators
  operate on binary numbers.
  used in low-level programming & networking

```
a = 5
b = 3
print (a & b)
```

6. Design and implement a menu-driven python program that

- Accepts user input
- Performs arithematic, relational, logical and assignment operatons.
- Displays results clearly

the program should handle int, float, and boolean data types.

```
a= int (input ("Enter first number : "))
b = int (input ("Enter second number : "))

print (" 1. Arithemtic ")
print (" 2. Relational ")
print (" 3. Logical ")
print (" 4. Assignment ")

ch = int(input (" Enter choice: "))

if ch = = 1:
    print ("Add = ", a+b)
elif ch = = 2:
    print("a > b = ", a > b)
elif ch = = 3:
    print ((a>0) and (b>0))
elif ch = = 4:
    a += b
    print( "a = ", a)
else:
    print ("Invalid choice")
```

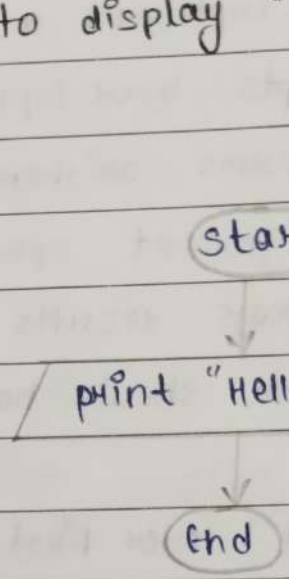
7. Write a program in python to display "Hello World"

Code

```
print ("Hello World")
```

Output

Hello World



Algorithm

1. Start
2. Print Hello World
3. Stop

8. Write a program to perform average of 5 numbers using input () function and type conversion concept.

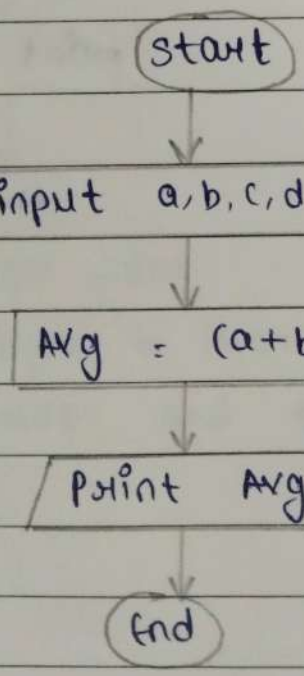
```
a= Float (input ("Enter number 1 : "))
b= Float(input ("Enter number 2 : "))
c= Float(input ("Enter number 3 : "))
d= Float (input ("Enter number 4 : "))
e= Float(input("Enter number 5 : "))

Avg = (a+b+c+d+e)/5
print ("Average = ", avg)
```

Output

Enter number 1 : 10
Enter number 2 : 20
Enter number 3 : 30
Enter number 4 : 40
Enter number 5 : 50
Average = 30.0

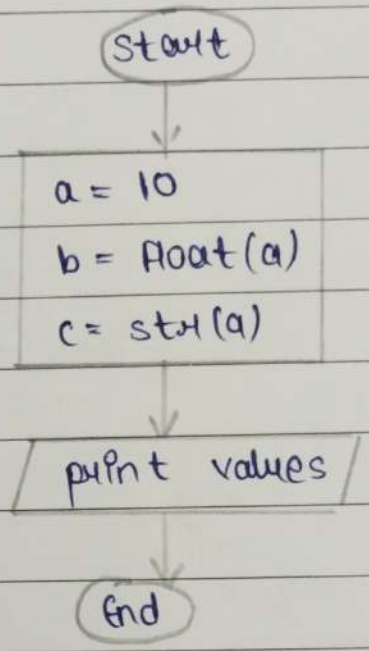
Algorithm.

1. Start
2. Accept 5 numbers from user
3. Calculate Average = (sum of numbers)/5
4. Display Average
5. Stop



10. write a program to demonstrate the concept of data types & conversion.

```
a=10
b= float (a)
c= str (a)

print (a, type(a))
print (b, type (b))
print (c, type (c))
```



Output

```
10 < class 'int' >
10.0 < class 'float' >
10 < class 'str' >
```

Algorithm

1. Start
2. Assign integer value
3. convert integer to float and string
4. Display values and their data types
5. stop